

PUKAPUKA, Cook Islands

WMO: 918110

Lat: 10.915S

Lon: 165.839W

Elev: 5

StdP: 101.27

Time Zone: -10.00 (W10)

Period: 86-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.9	24.6	21.0	15.7	27.4	21.9	16.6	27.4	11.5	26.7	10.3	27.6	2.8	90	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	4.4	32.5	27.2	32.1	27.0	31.8	26.9	28.2	31.0	27.9	30.9	27.7	30.6	3.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	23.3	30.2	27.1	22.9	30.0	26.8	22.5	29.7	90.9	31.4	89.4	31.0	88.0	30.6	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.1	9.2	8.6	17.1	33.5	1.8	0.7	15.8	34.0	14.7	34.3	13.7	34.7	12.4	35.2		
			15.6	28.8	2.6	0.8	13.7	29.4	12.1	29.8	10.6	30.2	8.7	30.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	28.7	28.5	28.6	28.9	28.9	28.5	28.0	28.1	28.3	28.5	28.5	28.4
	DBStd	1.09	0.96	1.14	1.14	0.99	1.08	1.06	1.02	0.91	1.06	1.03	1.09	1.19
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6745	580	518	576	567	585	555	558	561	549	572	554	569
	CDD18.3	3703	322	284	317	317	327	305	299	303	299	314	304	311
	CDH23.3	42233	3719	3242	3631	3731	3881	3523	3301	3318	3353	3561	3473	3499
	CDH26.7	14241	1316	1113	1281	1406	1461	1194	954	951	1052	1182	1182	1150
Wind	WSAvg	4.2	3.4	3.8	3.6	3.2	4.1	5.1	5.1	5.2	5.0	4.1	3.6	3.9
Precipitation	PrecAvg	2788	355	318	262	147	190	180	163	154	175	194	270	372
	PrecMax	3910	861	686	579	393	448	373	496	385	386	487	499	756
	PrecMin	1950	132	135	78	13	41	44	13	8	10	43	108	79
	PrecStd	487	181	160	135	83	91	86	117	89	84	100	98	151
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.6	32.6	32.7	32.7	32.5	31.9	31.5	31.5	32.1	32.4	32.5	32.5
		MCWB	27.0	27.1	27.4	27.7	27.4	27.0	26.2	26.3	26.6	26.6	27.1	27.3
	2%	DB	32.1	32.1	32.1	32.2	32.0	31.3	30.9	30.8	31.4	31.8	31.9	31.9
		MCWB	26.8	27.0	27.2	27.4	27.2	26.8	26.3	26.2	26.5	26.5	26.8	27.0
	5%	DB	31.6	31.5	31.6	31.7	31.5	30.8	30.4	30.4	30.9	31.2	31.4	31.3
		MCWB	26.7	26.8	27.1	27.2	27.1	26.6	26.2	26.1	26.3	26.4	26.7	26.9
	10%	DB	31.1	30.9	31.0	31.1	31.0	30.4	29.9	29.8	30.3	30.6	30.8	30.7
		MCWB	26.7	26.7	26.9	27.0	27.0	26.5	26.1	26.0	26.2	26.3	26.5	26.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.0	28.1	28.8	28.8	28.6	28.1	27.5	27.4	27.5	27.7	28.1	28.2
		MCDB	31.0	31.4	31.2	31.6	30.9	30.4	29.9	29.8	30.4	30.3	31.1	31.2
	2%	WB	27.6	27.6	28.0	28.1	28.1	27.6	27.1	27.0	27.1	27.2	27.5	27.7
		MCDB	30.6	30.9	31.1	31.1	30.9	30.0	29.6	29.7	30.2	30.1	30.6	30.7
	5%	WB	27.2	27.2	27.6	27.6	27.7	27.2	26.7	26.7	26.7	27.0	27.1	27.3
		MCDB	30.3	30.4	30.6	30.6	30.5	29.7	29.3	29.3	29.8	30.0	30.2	30.2
	10%	WB	27.0	26.9	27.1	27.2	27.3	26.9	26.5	26.3	26.5	26.7	26.8	27.0
		MCDB	30.1	30.0	30.2	30.1	30.0	29.4	28.9	28.9	29.4	29.6	29.7	29.8
Mean Daily Temperature Range	5% DB	MDBR	4.6	4.6	4.8	4.6	4.4	3.8	3.9	3.9	4.1	4.3	4.4	4.3
		MCDBR	5.3	5.4	5.5	5.3	5.0	4.4	4.7	4.5	4.8	5.2	5.2	5.1
	5% WB	MCWBR	2.2	2.3	2.5	2.2	2.4	2.3	2.3	2.2	2.3	2.3	2.4	2.3
		MCWBR	4.7	4.6	4.9	4.8	4.4	3.8	3.9	4.1	4.2	4.4	4.7	4.6
Clear-Sky Solar Irradiance	taub	0.387	0.392	0.379	0.382	0.386	0.392	0.389	0.395	0.398	0.382	0.382	0.383	
	taud	2.629	2.608	2.656	2.653	2.636	2.597	2.598	2.576	2.568	2.625	2.639	2.642	
	Ebn at Noon	954	944	940	907	873	851	862	882	908	942	953	957	
	Edh at Noon	101	103	96	92	89	90	91	98	103	100	99	99	
All-Sky Solar Radiation	RadAvg	5.58	5.66	5.75	5.57	4.95	4.71	5.00	5.46	6.11	6.18	5.98	5.91	
	RadStd	0.55	0.78	0.45	0.44	0.30	0.25	0.33	0.34	0.38	0.43	0.47	0.50	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page